HOMEWORK 2
DUE APRIL 27, 2026

Full credit (15 points) can be obtained by adequately answering 6 of the following questions. 5 extra credit points are available for exceptional additional effort. Acknowledge any collaboration and LLM use.

Exercise 0.1. Draw some pictures/diagrams illustrating some of the ideas from the course so far.

Definition. An **atom** of a boolean algebra $\mathcal{C}$ is a minimal element (with respect to inclusion) of $\{A \in \mathcal{C} : A \neq \varnothing\}$.

Exercise 0.2. Suppose X is a set, and $\mathcal{C} \subseteq \mathcal{P}(X)$ has cardinality $|\mathcal{C}| = n$. Give an upper bound on the number of atoms in $\mathcal{C}^{\text{bool}}$ and the cardinality of $\mathcal{C}^{\text{bool}}$.

Exercise 0.3. How many boolean algebras are there on a set of size n?

Exercise 0.4. Suppose $X \subseteq \mathbb{R}^2$ is semialgebraic. Suppose $X \cap \Gamma(\exp)$ is infinite, where $\Gamma(\exp)$ is the graph of the total real exponential function $\exp : \mathbb{R} \to \mathbb{R}$. Prove that X has nonempty interior.

Exercise 0.5. Suppose $A \subseteq \mathbb{R} \times \mathbb{R}$ is a symmetric and reflexive binary relation definable in a structure $\mathcal{R}$ on $\mathbb{R}$. Is the transitive closure A^{tr} of A necessarily also definable in $\mathcal{R}$? In other words, prove this is true in every structure $\mathcal{R}$, or find a structure $\mathcal{R}$ and a specific set $A \subseteq \mathbb{R}^2$ definable in $\mathcal{R}$ for which this is false.

Exercise 0.6. Suppose $\mathcal{R}$ is an o-minimal structure on $\mathbb{R}$ (satisfying (S5) and (S6)). Let $f : A \to B$ be a definable surjection, with $A \subseteq \mathbb{R}^m$ and $B \subseteq \mathbb{R}^n$ definable. Show there exists a definable function $h : B \to A$ such that $g \circ h = \mathrm{id}_B$.

Definition. (cf. [1, Chapter 4]) Suppose $C : (0, +\infty) \rightrightarrows \mathbb{R}^n$ is a set-valued function. Define the **outer limit** (of C, as $t \downarrow 0$):

$$\begin{aligned}\limsup\nolimits_{t\downarrow 0} C_t \;:=\; \{&x \in \mathbb{R}^n : \text{for every neighbourhood } U \text{ of } x, \text{ and every } t' > 0,\\ &\text{there exists } 0 < t'' < t' \text{ such that } C_{t''} \cap U \neq \varnothing\}\end{aligned}$$

Likewise, define the **inner limit**:

$$\begin{aligned}\liminf\nolimits_{t\downarrow 0} C_t \;:=\; \{&x \in \mathbb{R}^n : \text{for every neighbourhood } U \text{ of } x, \text{ there exists } t' > 0 \text{ such that}\\ &C_{t''} \cap U \neq \varnothing \text{ for every } 0 < t'' < t'\}\end{aligned}$$

If the outer limit and the inner limit agree, then the common set is called the **Painlevé-Kuratowski** limit, denoted:

$$\lim\nolimits_{t\downarrow 0} C_t \;:=\; \limsup\nolimits_{t\downarrow 0} C_t \;=\; \liminf\nolimits_{t\downarrow 0} C_t$$

Exercise 0.7. Suppose $\mathcal{R} = (\mathbb{R}; <, \ldots)$ is a structure satisfying (S1)-(S4) and (O1), and suppose $C : (0, +\infty) \rightrightarrows \mathbb{R}^n$ is definable in $\mathcal{R}$.

(1) Show that $\limsup_{t\downarrow 0} C_t$ and $\liminf_{t\downarrow 0} C_t$ are both definable in $\mathcal{R}$.
(2) Now assume $\mathcal{R}$ is an o-minimal structure additionally satisfying (O2), (S5), and (S6). Prove that $\limsup_{t\downarrow 0} C_t = \liminf_{t\downarrow 0} C_t$.
(3) Give a (non-o-minimal) example where $\limsup_{t\downarrow 0} C_t \neq \liminf_{t\downarrow 0} C_t$.

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Exercise 0.8. (cf. [1, Chapter 6]) Given a set $C \subseteq \mathbb{R}^n$, define the **tangent bundle** T_C of C to be the following set-valued map:

$$T_C \;:\; \mathbb{R}^n \rightrightarrows \mathbb{R}^n, \quad x \;\mapsto\; T_C(x) \;:=\; \limsup_{\tau \downarrow 0} \tau^{-1}(C - x)$$

The set $T_C(x)$ is called the **tangent cone** of C at x. Likewise, we also define the **derivable tangent bundle** T_C^{der} of C to be the following set-valued map:

$$T_C^{\mathrm{der}} \;:\; \mathbb{R}^n \rightrightarrows \mathbb{R}, \quad x \;\mapsto\; T_C^{\mathrm{der}}(x) \;:=\; \liminf_{\tau \downarrow 0} \tau^{-1}(C - x)$$

and we call $T_C^{\mathrm{der}}(x)$ the **derivable tangent cone** of C at x.

Exercise 0.9. Suppose $\mathcal{R} = (\mathbb{R}; <, +, \cdot, \ldots)$ is a structure on $\mathbb{R}$ satisfying (S1)-(S4), (O1) and (S5)-(S6), and suppose $C \subseteq \mathbb{R}^n$ is definable in $\mathcal{R}$.

(1) Show that T_C and T_C^{der} are both definable in $\mathcal{R}$.
(2) Now assume $\mathcal{R}$ is additionally an o-minimal structure, i.e., also satisfies (O2). Prove that $T_C(x) = T_C^{\mathrm{der}}(x)$ for every $x \in \mathbb{R}^n$.
(3) Give a (non-o-minimal) example where $T_C(x) \neq T_C^{\mathrm{der}}(x)$.

Exercise 0.10. Suppose $I \subseteq \mathbb{R}$ is an open interval, and $f : I \to \mathbb{R}$ is strictly increasing. Further assume that $J \subseteq \mathbb{R}$ is an open interval such that $J \subseteq f[I]$. Show the following:

(1) $K := f^{-1}(J) \subseteq \mathbb{R}$ is an open interval, and
(2) $f|_K : K \to \mathbb{R}$ is continuous.

Exercise 0.11. Suppose f is definable and $I \subseteq \mathbb{R}$ is an open interval. If $f : I \to \mathbb{R}$ is continuous and Fréchet-differentiable, then f is C^1-smooth (i.e., Fréchet-differentiable with continuous derivative). Is this still true if f is not definable?

Exercise 0.12. Suppose $C \subseteq \mathbb{R}^n$ is convex. Prove that C is *semi-closed*, i.e., $\mathrm{int}(C) = \mathrm{int}(\mathrm{cl}(C))$. Show that the boundary of a convex set is nowhere dense.

References

1. R Tyrrell Rockafellar and Roger J-B Wets, *Variational analysis*, vol. 317, Springer Science & Business Media, 2009.